Activity: Constraint Satisfaction Problems
For Sections: B, E, F

Part 1: Course Scheduling

You are scheduling 5 computer science classes across 3 professors (A, B, and C).

The Classes:

- **C1:** Intro to Programming (8:00–9:00am)
- **C2:** Intro to AI (8:30–9:30am)
- **C3:** NLP (9:00–10:00am)
- **C4:** Computer Vision (9:00–10:00am)
- **C5:** Machine Learning (9:30–10:30am)

The Professors:

- **A:** available to teach Classes 3 and 4.
- **B:** available to teach Classes 2, 3, 4, and 5.
- **C:** available to teach Classes 1, 2, 3, 4, 5.

1.1 Problem Formulation (2 Marks)

Formulate this as a CSP by defining the variables, their initial domains (based on professor availability), and the binary constraints (based on time overlaps).

1.2 Constraint Graph (2 Marks)

Draw the constraint graph associated with your CSP. Nodes represent classes, and edges represent a conflict (overlap) where the same professor cannot be assigned.

1.3 Arc-Consistency (4 Marks)

Show the domains of the variables after running **Arc-Consistency (AC-3)** on the initial graph.

Variable	Initial Domain	Final Domain (After Arc-Consistency)
C_1	$\{C\}$	
C_2	$\{B, C\}$	
C_3	$\{A, B, C\}$	
C_4	$\{A, B, C\}$	
C_5	$\{B, C\}$	

1.4 Solution (2 Marks)

Provide **one** complete valid solution (assignment of professors to classes).

$$C_1 = \_\_\_\_\_,\ C_2 = \_\_\_\_\_,\ C_3 = \_\_\_\_\_,\ C_4 = \_\_\_\_\_,\ C_5 = \_\_\_\_\_$$

Part 2: Conceptual Logic (5 Marks)

1. If Class 4 were taught only by Professor A, how would this "unary constraint" affect the domain of Class 3 during the arc-consistency check?

2. Would **Forward Checking** be more or less efficient than **Arc-Consistency** if we were solving this for a university with 500 classes? Explain briefly.

Part 3: Cryptarithmetic (10 Marks)

Instructions: In the following puzzles, each letter represents a **unique decimal digit** ($0-9$). The leading digit of any multi-digit number cannot be zero. **Each problem is a separate question with separate solutions.**
Solve by hand and show complete working.

2.1 (2 Marks)

$$
\begin{array}{cccccc}
 & C & R & A & C & K \\
+ & & H & A & C & K \\
\hline
 & E & R & R & O & R
\end{array}
$$

2.2 (2 Marks)

$$
\begin{array}{cccccc}
 & B & L & A & C & K \\
+ & G & R & E & E & N \\
\hline
O & R & A & N & G & E
\end{array}
$$

2.3 (2 Marks)

$$
\begin{array}{ccccc}
 & & E & A & T \\
+ & T & H & A & T \\
\hline
A & P & P & L & E
\end{array}
$$

2.4 (2 Marks)

$$
\begin{array}{ccccc}
 & F & O & U & R \\
+ & F & I & V & E \\
\hline
 & N & I & N & E
\end{array}
$$

2.5 (2 Marks)

$$
\begin{array}{cccccc}
 & C & R & O & S & S \\
+ & R & O & A & D & S \\
\hline
D & A & N & G & E & R
\end{array}
$$